UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Abstract

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The study of Higgs properties in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \to 4\ell$ decays ($\ell = \mathrm{e}$, μ) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV . The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1}at 13 TeV(Full Run 2 data). These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})\ \mathrm{fb}$ at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. These measurements are important to test the Standard Model predictions in the Higgs sector and can be used to constrain phase spaces of many theories beyond the Standard Model.

All the integrated and differential measurements are compared with the Standard Model based theoretical predictions and found to be consistent within their uncertainties.

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Chapter 1

Introduction

After a quest, ordinary matter and underlying forces or interactions (electromagnetic, weak and strong interactions and not the gravity) which bind it in a structure are successfully explained by the Standard Model (SM) of the particle physics. In SM, description of these interactions is made by symmetry laws of mathematics which could only explain the characteristic of particles with zero mass. Higgs mechanism was then introduced to make the theory realistic enough to encorporate mass problem. Under this mechanism, electroweak symmetry breaks and as a result particles gain the mass. Higgs boson is the direct manifestation of this Higgs mechanism [1, 2, 3, 4, 5, 6].

Precision measurements on the high energy physics experiments like Tevatron and LEP have proven the authenticity of SM; however, Higgs observation could not happen. In 2010, Large Hadron Collider (LHC), the largest experiment ever of its kind in physics world, was then set to put more tests on SM and it endorsed the measurements on SM from earlier experiments. In 2012, the "hidden" Higgs boson, was discovered in shape of heavy scalar boson as jointly reported by the CMS and ATLAS experiments [7, 8]. With this happening, physicists were all set to measure other properties of Higgs for further tests on SM predictions. Mass of Higgs boson was only free parameter in Higgs sector of SM until measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [9].

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \to 4\ell$ decay channel ($\ell = e, \mu$) using the full Run 2 CMS data recorded in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. Included Higgs differential measurements are based on Higgs kinematics and motivated from several aspects. For instance, Higgs trasverse momentum can be used to test the pQCD calculations (especially when measured with exclusive multiplicities of jets). Additionally this variable is sensitive to Lagrangian structure of Higgs. Rapidity of Higgs is sensitive to parton distribution functions (PDFs) of the protons involved in the collisions. Jet related observables which are sensitive to theoretical modelling, boosted quark and also to the gluon emission are also covered in the dissertation. Current measurements are consistent with SM prediciton; however, in such measurements, any deviation from SM expectations provides a direct hint to BSM contributions. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different decay cahannels of Higgs which can been studied in Ref. [**Khachatryan:2015yvw**, **CMS:2015hja**, **CMS:2016rqf**, **Sirunyan:2017exp**, **CMS:2018hhg**, **atlas-conf-2019-029**, **atlas-conf-2019-025** , **atlas-conf-2019-032**, **CMS:2019drq**, 10, 11, 12, 13, 14, 15, 16]. More precise Higgs boson properties measurements are needed to spot any possible deviation from SM predictions which can be a doorway to new physics.

This thesis is structured as followed:

- **Chapter 1 ??:** An overview of theoretical background i.e. the SM of particle physics, production and decays of H boson are described.
- **Chapter 2:** Overview of LHC experiment and its operation are described.
- **Chapter 3:** CMS detector, its sub-detectors and gradual upgrades are described.
- **Chapter 4 ??:** The calibrations and reconstructions of the physics objects within the CMS detector are described.
- **Chapter 5 ??:** Presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV.
- **Chapter 4:** Finally, the conclusions and outlook of the dissertation are summarized.

Chapter 2

The Large Hadron Collider (LHC)

LHC is located at CERN (European Organization for Nuclear Research) located at Franco-Swiss border near Geneva, Switzerland and is most powerful setup of its kind to probe resarches in particle physics. From its design, it is capable enough to accelerate and then collide two proton beams [1] at center of mass energy upto 14 TeV at peak luminosity of $\mathscr{L} = 10^{34}\mathrm{cm}^2\mathrm{s}^{-1}$.It was constructed from 1998 to 2008, in a 3.8 meters wide tunnel of 27 Kilometers circumference at a depth which varies from 75-150 meters underground. This tunnel was previously used for LEP collider which was constructed from 1983 to 1988. Two counter rotating proton beams (anti-proton beam is disfavored to use) in two rings are used where the beam pipes are kept at ultra-high vacuum just to preserve the beams from possible extra collisions with molecules of gas. To orient the beams in alongside the tunnel superconducting (cryo) dipole magnets are used as sketched in 2.1.

2.1 LHC beam operations

Beams of protons are inserted to the LHC experiment at injector. Protons are obtained by removing electrons using high electric field. Injector is connected to Linac2, proton synchrotron booster, proton synchrotron and the super proton synchrotron booster (PSB) in which the proton beams pass and with a subsequent gain in energy at each step (50 MeV to 450 GeV from Linac2 to PSB). From the PSB, beam is then passed to LHC tunnel where beams accelerates until each beam attains the energy of 6.5 TeVfor the subsequent collisions to be recorded at dedicated experiments or detectors of LHC. The schematic diagram of LHC accelerating mechanism is given in the Fig. 2.2

[1] LHC also does collision experiments of Heavy(Pb) ions at 2.8 TeVper nucleon. Details are beyond the scope of the dissertaton

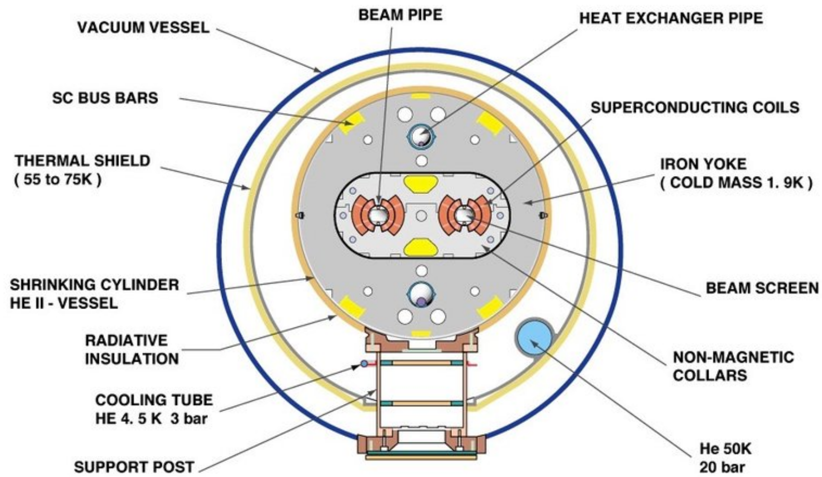


FIGURE 2.1: Cross section of the LHC cryodipole

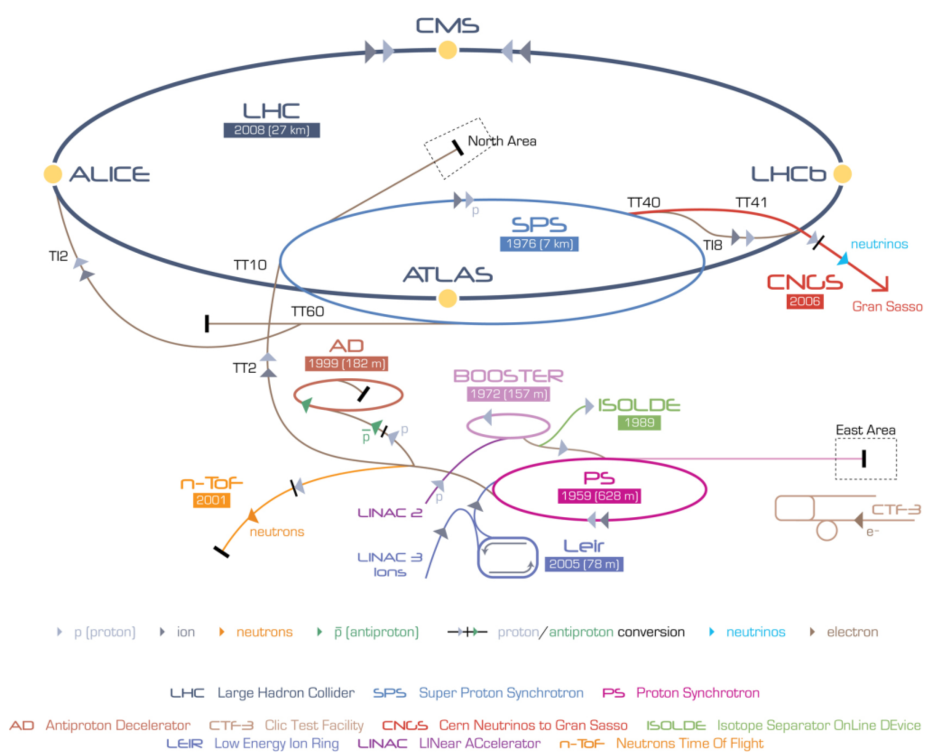


FIGURE 2.2: Accelerating mechanism of LHC experiment at CERN

For the first time, beam of protons was injected into LHC on September 10, 2008. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, due to an electric defect magnet quench was triggered about a hundred bending magnets resulting in several damages. After the repair, in November 2009, LHC became world's strongest particle accelerator after having achieved 1.18 TeV energy each beam thus defeating the Tevatron's highest achieved energy of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached center of mass energy up to 7 TeV (3.5 TeV each beam). In 2011 and then in 2012, LHC consistently delivered data at 7 and 8 TeV energies respectively. During 2013-14 , LHC machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV close to full stipulated operating energies (14 TeV). After successfully delivering proton-proton collisions data during Run 2 till the end of 2018, LHC machine is shutdown for Long Shutdown 2 (LS2) upgrades and is expected be operational again for Run3 in 2021 after several upgrades.

2.1.1 Machine Layout and Performance

Basic layout of LHC is shown in Figure 2.2 [17]. The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. The ATLAS and CMS, two general purpose detectors of LHC, are installed opposite to each other (at P1 and P5 respectively). The other two important detectors, ALICE and LHCb, are installed (at P2 and P8 respectively) as shown in the same Figure 2.2 which are designed to study heavy ion Physics and B Physics (b quark) respectively. Two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. In order to keep the beam alongside the tunnel during the process of accelerating, magnetic force is applied through diploe magnet made of $NbTi$ superconducting material. Apart from that, quadrupole magnet is used to focus the beam. Once each beam attains targeted energy i.e. 7 TeV, they are made collided at differents spots of LHC tunnel where different experiments are installed.

LHC design goal is to test SM Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. In collision process, number of events (N) per unit time (called event rate) for a certain process produced at LHC are given as

$$\frac{dN}{dt} = L \times \sigma_{process} \tag{2.1}$$

where $\sigma_{process}$ being the cross section of the process and L is the instantaneous machine luminosity. Integrating over time gives the total number of events in the particular process,

$$N = \int Ldt \times \sigma_{process} = \mathscr{L}\sigma_{process} \tag{2.2}$$

Here $\mathscr{L}$ represents *integrated* machine luminosity. All possible processes' cross sections produced during proton-(anti) proton collisions at LHC machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$ are shown in Figure 2.3. It is obvious from the diagram that an increase in machine luminosity and center-of-mass energies would enable us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathscr{L}$ = 10^{34}cm^{2}s^{-1}, and nominal center of mass energy of LHC is $\sqrt{s} = 14$ TeV.

Nominal Luminosity in terms of Beam Parameters

The machine luminosity can be expressed in terms of beam parameters as:

$$\mathscr{L} = \frac{\gamma_r N_b^2 f_{rev} k_b}{4\pi\epsilon_n\beta^*}F \tag{2.3}$$

here γ_r represents relativistic factor; however, N_b is bunch intensity, k_b being number of bunches in a beam; f_{rev} is the revolution frequency, β^* represents the beta function at the collisions point and F is the luminosity depletion factor due to cross angle at interaction point (IP). This factor can be written as below

$$F = 1 + (\frac{\theta_c\theta_X}{2\sigma^*})^2 \tag{2.4}$$

where θ_c, θ_z and σ^* are crossing angle at point of interaction, root mean square (RMS) length of bunch and σ^* the transverse RMS size of beam at IP. In above expression of luminosity, some parameters can be used to increased the luminosity at LHC; however, some of them cannot be used e.g. f_{rev} which is fixed by perimeter

TABLE 2.1: Parameters' values for LHC operations.

Parameter	Value
Circumference	26659 m
Center of mass energy ($\sqrt{s}$)	14 TeV
Momentum at collision	7 TeV/s
Luminosity (nominal)	$10^{34} cm^2 s^{-1}$
Bunch spacing	25 ns
Minimum distance between bunches	7 m
No. of bunches in ring	2808
No. of protons in a bunch	1.15×10^{11}
Beta function at the collision point(β^*)	0.55 m
Transverse RMS size of beam at impact point(σ^*)	16.3 μm
Peak magnetic field of dipole	8.33 T
Beam lifetime	10 h
Current in main dipole	11800 A
Stored energy in magnets	11 GJ
No.of dipoles	1232
No.of quadrupoles	858
Minimum distance between bunches	7 m
Operating temperature of dipole magnet	1.9 K

of LHC tunnel and protons' speed in the beam. The expression above is valid for two circular beams with same parameters.

The two major experiment of LHC, the CMS [18] and the ATLAS [19], are designed to operate at luminosity of $\mathscr{L} = 10^{34} cm^2 s^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 2.1.

By the end of the Run 2, LHC has delivered 150 $\mathrm{fb}^{-1}$of data. However, to fully explore the SM, particulary the Higgs sector, and even beyond SM for the rare processes, physicists are really in need of sufficient statistics (i.e. higher luminosity). High luminosity LHC is, therefore, planned to achieve a peak luminosity of 5 $\times 10^{34} cm^2 s^{-1}$ - 10 $\times 10^{34} cm^2 s^{-1}$.

2.2 Data Taking from 2015 to 2018 (Full Run 2)

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design

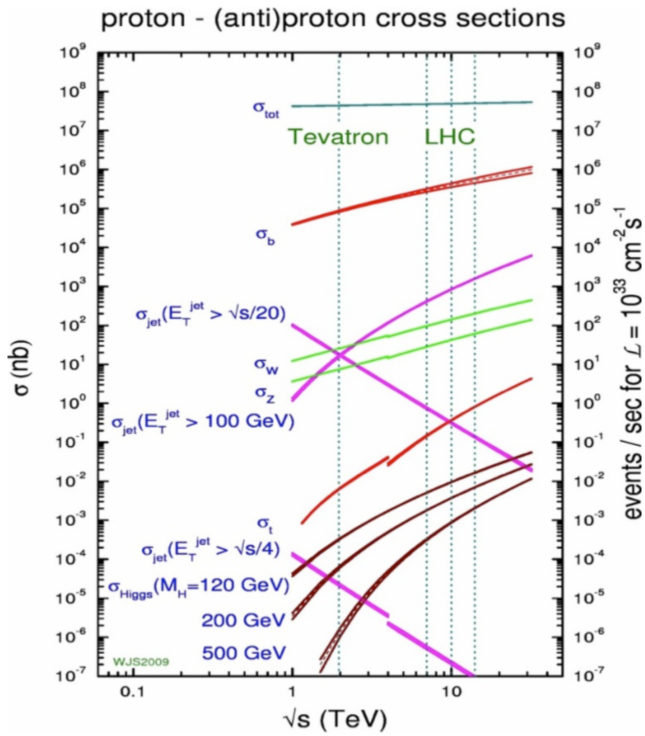


FIGURE 2.3: Cross section values for different process corresponding to machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$.

value i.e 14 TeV). LHC delivered 4.2 fb^{-1}during 2015 at 13 TeV center of mass energy and subsequently, it delivered 41.0 fb^{-1}, 49.8 $\mathrm{fb}^{-1}$and 67.9 fb^{-1}during the years 2016, 2017 and 2018 as shown in the Figure 2.4. In the figures we can also see the recorded data by the CMS experiment from LHC delivered data in different run periods.

2.3 LHC upgradation during Long shutdown 2 (LS2)

After successfully delivering data during its Run2 as described in section 2.2, LHC is passing through its long shutdown 2 period. During this period, several upgardes are ongoing which can be regarded as preparations for High Luminosity LHC later on (HL-LHC). However, main upgrades in this regards will take place during LS3 after Run 3 period of LHC. One is the replacement of Linac2 with the Linac4. It is about 90 meter in length that took around 10 years to build it.It would be capable to accelerate negative hydrogen ions (H^-) for beam energy upto 160 MeV and will enable beam to double its intensity Ref. **??**. Besides, the individual

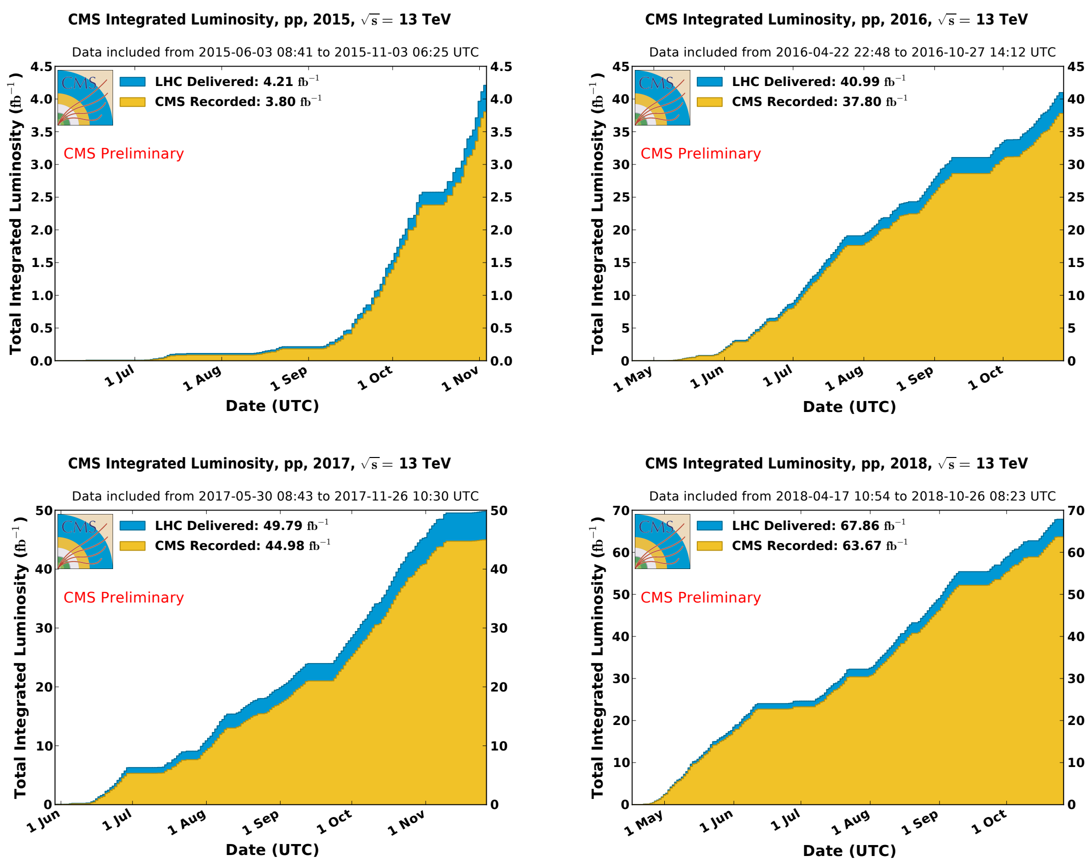


FIGURE 2.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for year 2015 (top left), year 2016 (top right), year 2017 (bottom left) and year 2018 (bottom right).

experiments of LHC, i.e. CMS, ATLAS, ALICE and LHCb, have also planned certain upgarades during this LS2 period Ref. **??**.

ATLAS is builing muon small wheels to be integrated with the trigger system. For high pileup LAr electronics are being upgraded.

CMS is changing the photodetectors (HPD) of barrel and endcap HCAL to the silicon photo-multipliers (SiPM). Also the beam pipes made with stainless steel are being replaced with Aluminium composed material. Also, based on damage from radiation in endcaps of HCAL, scintillating tiles would be changed. Furthermore, for HL-LHC preprations, CSC muon chambers are being upgraded with new front end (FE) elecronics.

ALICE is making several upgrades for Inner Tracking System (ITS), Time Projection Chamber (TPS), Central Trigger Processor, Data Acquisition (DAQ) High Level Trigger (HLT), Muon Forward Tracker (MFT) and so on.

LHCb is improving its readout from 1 MHz to 40 MHz with trigger based fully on softwares by replaceing FE electronics and new computing infrastructure.

Chapter 3

The Compat Muon Solenoid (CMS) detector

Since the data used for the analysis in the dissertation was collected using the CMS detector, this chapter would focus on geometry of CMS, its parts (sub-detectors) and description of how events are reconstructed with the CMS detector.

3.1 The Compact Muon Solenoid (CMS) detector

LHC has set new frontiers for researches in particle physics experiments. State of the art experimental facility has annexed with equipment based on latest technology. To assure the validity of results from the experiment, LHC has its two major and general purpose experiments for cross-validation, one of which is CMS. They have same physics goals with some different design and the detector technology. Particulary the CMS is capable to provide good resolution for muon identification and momentum over wide range of dimensions and determination of the charge on muons with ultimate precision with momentum up to 1 TeV. It gives healthy inner tacker reconstruction efficiency and resolution of charged particle momentum. It has efficient trigger system and better discrimination of τ's and b-jets. Further, it is benefitted from good dielectron, diphoton and dijet mass and energy resolutions, and as well as missing-transverse-energy (MET) resolutions. This makes it capable to study large areas from SM Physics (including Higgs Physics) to search for extra dimensions and the dark matter covering physics in energy range from a few 100 GeV to TeV scale [20].

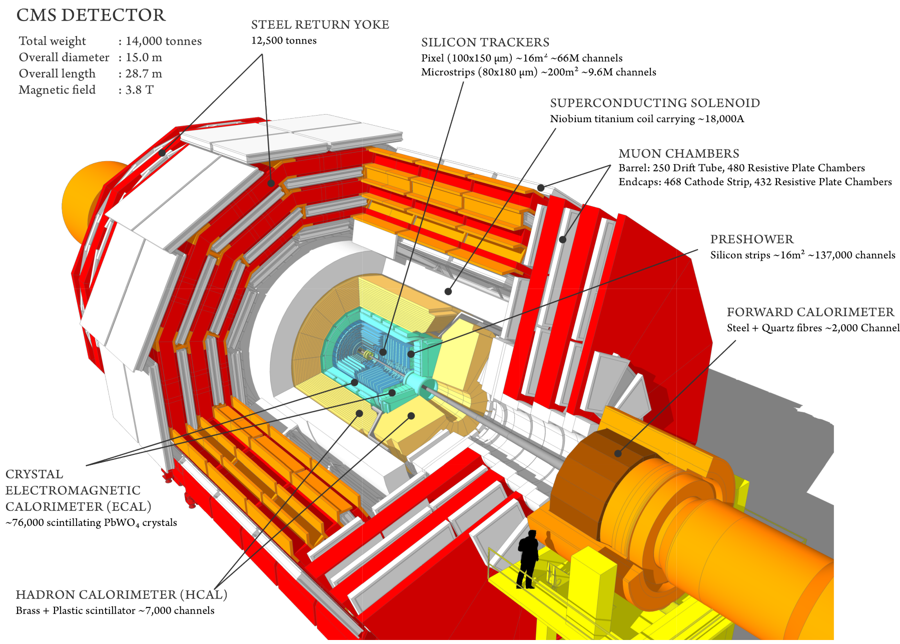


FIGURE 3.1: The CMS detector schematic diagram

3.1.1 General aspects and the geometry

Being one of the two general purpose detectors installed at LHC ring, CMS is is constructed in cylinderical shape having length about 28 m and outer ratdius of around 7 m. Its overall height is about 15 m with overall weight of about 14,000 tonnes. Overall deisgn of the CMS detector can been seen in the Figure 3.1. Learning from the earlier experiment at CERN, CMS engineered to be built in different slices at surface instead of doing everything underground. And fifteen separate slices were then lowered to underground to be fixed at the LHC tunnel.

Moving radially outwards from interaction point of beams, there are inner tracker system (ITS), electromagnetic calorimeter (ECAL) and hadronic calorimeter(HCAL) sub-detectors of CMS which are fitted inside the large solenoid magnet. Inner tracker of CMS have large track multiplicity as being closest to beam pipe where collisions happens. ECAL is build with crystals of lead tungstate which covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Outermost layer of the CMS detector is the Muon System. Details of sub-detectors, trigger system and data acquisition systems will be described in next Sections.

The CMS magnet

Measuring the momentum of the charged particle with great precision is one the ultimate goals of the CMS detector. In this regards, choice of magnetic field configuration is very important feature of a particle detector like CMS. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting the superconducting solenoid magnet.

When a charged particle moves in uniform magnetic field of a solenoid, its momentum resolution which depends upon strength of magnetic field and the radius of solenoid, can be written as:

$$\frac{dp}{p} \propto \frac{p}{Br^2} \tag{3.1}$$

Here r is the path length traversed by the charged particle inside the magnetic field (it becomes radius of the magnetic if the particle passed the full magnet). It is obvious from the relation, that for detector to obtain high momentum resolution, we need a volumetric solenoid magnet with stronger magnetic field. CMS solenoid magnet is designed to have bore of diameter 6 m and length 12.5 m producing magnetic field of 3.8 T which is 100,000 time stronger than earth magnet. It is the largest ever built magnet weighs about 12,000 tonnes. It is powerful enough in providing a strong magnetic field in its dimensions throughout the detector. It surrounds CMS ITS, ECAL and HCAL. The strong magnetic field efficiently bends the charged particles when they move through it. More the momentum of the charged particles less they bend, so high energetic charged particles require a stronger magnetic field for its momentum to be measured accurately. CMS aims to have excellent momentum resolution with such strong magnetic field specially for muons. The schematic diagram of CMS magnet is given in the Figure 3.2.

The CMS magnetic is made of coil of NbTi wire which becomes a source of uniform magnetic field when current passes through it. The NbTi becomes superconductor at -268.5° C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons [21, 22, 23, 24].

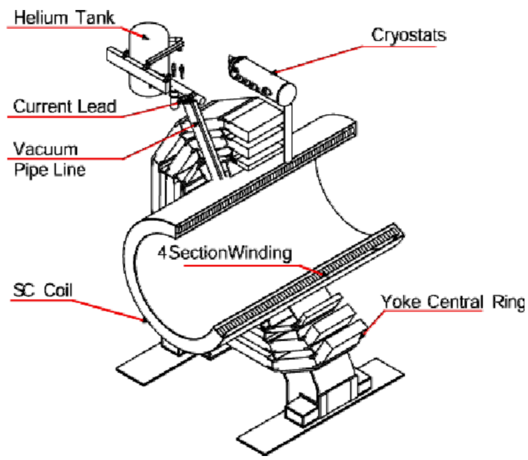
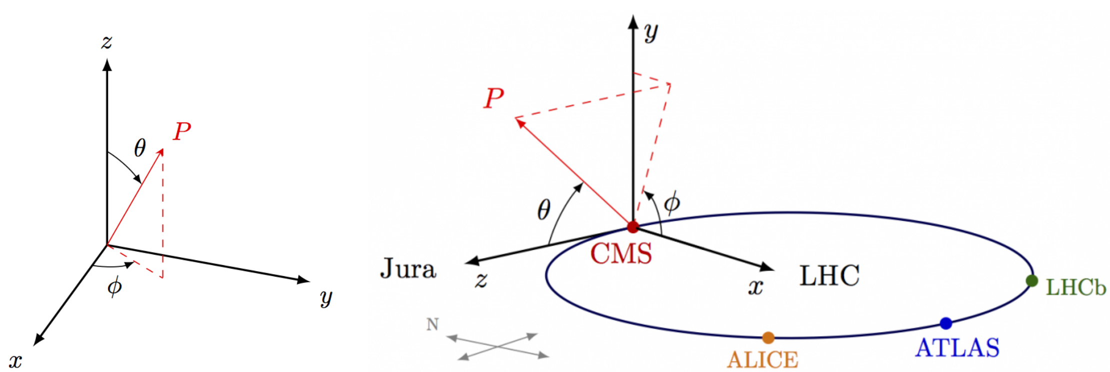


FIGURE 3.2: CMS magnet system.

3.1.2 CMS Coordinate system

Coordinate system used for CMS detector for its operations is sketched in Figures 3.3(a) and 3.3(b). The CMS detector is in cylinderical shape around beam axis as illustrated in Figure 3.1 which is represented by $Z - axis$ in Figures 3.3(b). The point where two beams collides in beam pipe is called center of experiment; the y axis is directed vertically upwards and the x axis being horizontal towards the LHC ring center.



(a) (b)

FIGURE 3.3: CMS spherical coordinte system

The r represents the radial component in transverse plane while ϕ is the the azimuthal angle measured from the x axis. θ which is measured from z axis is called polar angle. Position of particle when it passes through the CMS is described in terms of (η, ϕ), where η being the pseudo-rapidity defined as $\eta = -\ln\tan(\theta/2)$ and remains invariant under the longitudinal boosts in the environment of the partons which carry the fractions of the momentum from hard scattering collisions of e.g. proton-proton. Moreover, pseudorapidity also helps to differentiate the barrel regions from the endcap regions of sub detectors. In terms of pseudorapidity, the barrel region are defined by $|\eta| < 1.5$ whereas the endcap region corresponds to $1.5 < |\eta| < 3.0$.

3.1.3 Inner Tracking System (ITS)

The CMS ITS is a sub-detecting system of the CMS detector which is important form its task to give best tracks and hence the momenta (above 1 GeV) of the particles (electron, muon, hadrons-quark-structured particles and other short-lived particles) produced in the collisions that take place at CMS, LHC. It is the most inner sub-detector placed under influence of a strong and uniform magnetic field of CMS solenoid and it surrounds the region of beam pipe as shown in Figure 3.1. ITS also provides the position information of the primary and secondary vertices which is important to reject extra interactions and objects.

Being in cyliderical shape, it as diameter of about 2.5 m and length about 5.5 m providing $|\eta|$ coverage up to 2.5. At LHC nominal luminosity of $10^{34} cm^2 s^{-1}$, there is an average of more than 20 proton-proton interactions (called pileup, for 2018 it was above 30) leading to average about 1000 particles emerging after every bunch crossing, 25 ns. ITS is entirely made up of (200 m^2 of) silicon which is benefitted from being super efficienct in the environment of huge flux of particles coming out of the interaction point. A schematic diagram of CMS ITS is shown in the figure 3.4 (as stood before phase 1 pixel upgrade).

To limit the damage mainly by radiation ITS is kept at about -20° C. Full ITS is an assembly of a pixel detector and silicon strip tracker.

The Pixel Detector

Part of the tracker system which is nearest to the interaction point is the Pixel detector. Silicon modules of the pixel detector, are facilitated with pixel cells of size $100 \times 150 \mu m^2$ each providing the $|\eta|$ range up to 2.5. It helps to reconstruct the secondary vertex and delivers three spatial points precisely throughout its geometry [20].

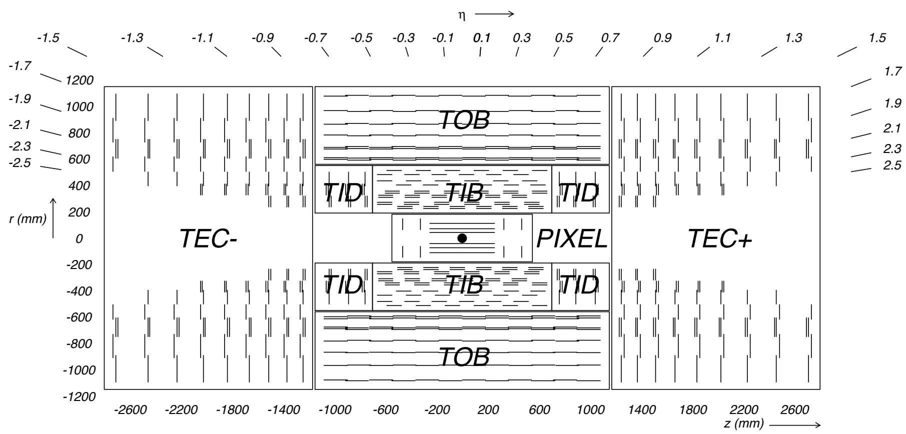


FIGURE 3.4: Schematic diagram of CMS tracker system. Lines represent detector modules (before phase 1 pixel upgrades).

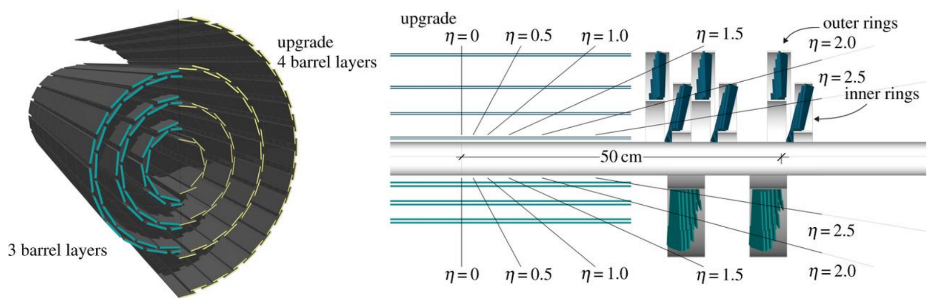


FIGURE 3.5: Schematic diagram of pixel detector's upgrade. Old pixel detector is shown by the lower half detector, and the new pixel detector is represented by the upper half.

During data taking of Run 1 operations of LHC, some dynamic inefficiencies (about 5% at an event pile-up of 40) were reported in the first layer of barrel of pixel sub-detector, due to the limited size of the readout bandwidth. Subsequently, a new pixel detector was installed in March 2017 Ref. **??**, equipped with one additional barrel layer and one additional forward disk. In the new pixel detector, the innermost layer is closer to the interaction point and the outermost one is further away from it as it is shown in Fig. 3.5

It also uses new Readout Chips (ROCs) benefitted from reduced data loss at higher collision rates with respect to Phase 1 operations (at 160 Mbit/s). Bandwidth of the readout electronics was increased approximately by a factor of eight (320 Mbit/s). In addition, DC-DC power converters were installed to allow reusing the existing fibers and cables. Hence, the upgraded pixel detector has a higher

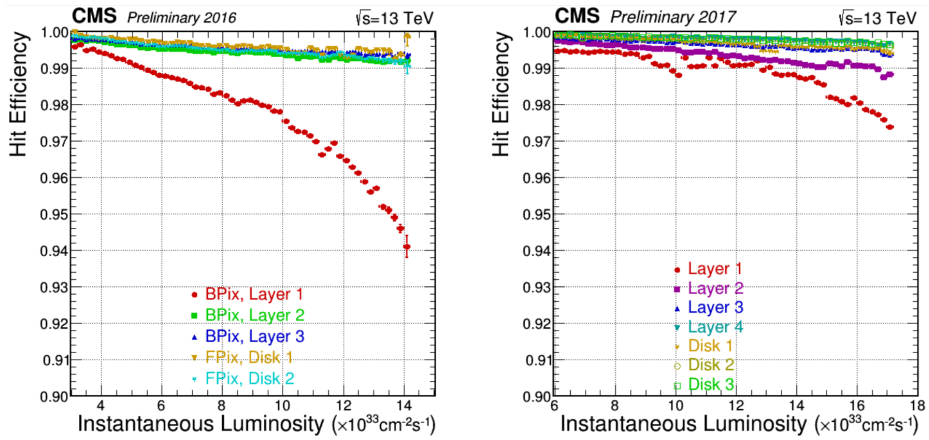


FIGURE 3.6: Pixel detector's hit efficiency as function of instantaneous luminosity corresponding to 2016 data (left) and to 2017 data (right)

rate capability and number of channels, providing increased tracking efficiency, reduced fake rate, and improved impact parameter resolution. In the figure 3.6, clear improvments in the dynamic efficiency of 2017 (with pixel upgrades) with respect to the 2016 (without pixel upgrades) can be seen Ref. **[Sonneveld:2018ntf]**.

Overall, the pixel detectors is of 1 m^2 size containing 66 million pixels.

The Silicon Strip

Next layer to the pixel detector is called Silicon Strip Tracker which is installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with Its tacker inner barrel (TIB, 4 layers) and tracker inner disk (TID, 3 layers) surrouds the pixel detector and extend upto 20 cm-50 cm where silicon microstrips of thickness 320μm are installed. Both are surrounded the Tracker outer barrel (TOB, 6 layers) which is 220 cm long and provides z coverage up to 118 cm. In barrel and endcap regions, the pitch of strips changes from 80μm to 120μm and 100μm to 141μm respectively. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing 500μm thick 6 barrel layers with strip pitch of 183μm and 122μm. Beyond z=118 cm, there are tracker endcaps (TEC+, TEC-, 9 discs) which cover 124 cm< $|z|$ <282 cm and 22.5 cm< $|r| < 113.5$ cm providing coverage of $|\eta|$<2.5.

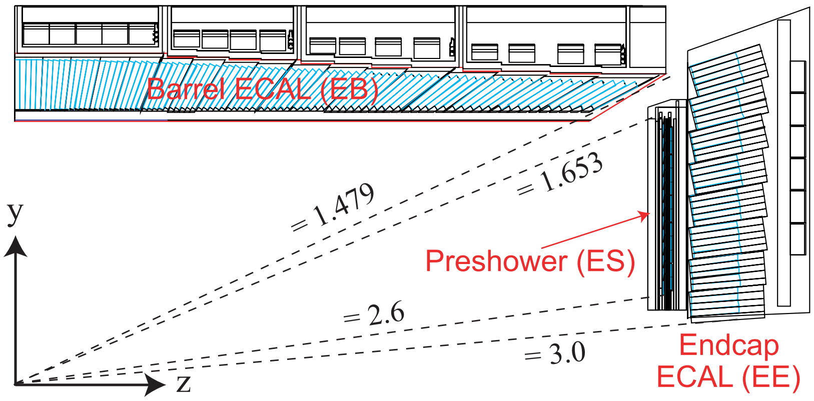


FIGURE 3.7: CMS ECAL pseudo-rapidity coverage.

3.1.4 Electromagnetic Calorimeter (ECAL)

ECAL is the cylindrical shaped homogeneous calorimeter built with 75848 crystals of lead tungstate ($PbWO_4$) placed inside the CMS solenoid magnet. Its barrel ergion (EB) contains 61200 $PbWO_4$ crystals and provides pseudorapidity coverage of $|\eta| < 1.479$. Endcap region of the ECAL (EE) is built with 7324 crystals on either side in endcap regions and provides pseudorapidity coverage up to $\eta < 3$. The region of $1.4442 < |\eta| < 1.56$ is regarded as crack region between barrel and endcaps as can been seen in Figure 3.7 taken from [20]. This region is excluded for physics analysis.

When high energy particle (e.g. electron or photon) coming out of the collision point enters ECAL, it produces bunch or cascade of particles (electron through bremsstrahlung and photon through pair production) with relatively lower energies. This is called EM showering that continues until electrons dominate in ionization and photon energy becomes less than pair-production threshold.

Photodetectors Avalanche photodiodes (APDs) in barrel region and the Vacuum phototriodes (VPTs) in endcap regions are installed which efficiently work under high magnetic field environment [25]. Response of ECAL comes with emission of light that is directly proportional incident particle's energy (through ionization).

A preshower detector is installed in endcap regions to distinguish between single photon and closeby diphoton candidate e.g neutral pion decay ($\pi^\circ \rightarrow \gamma\gamma$). Providing pseudorapidity coverage of $1.653 < |\eta| < 2.6$, it helps to efficiently determin the position of an electron and a photon.

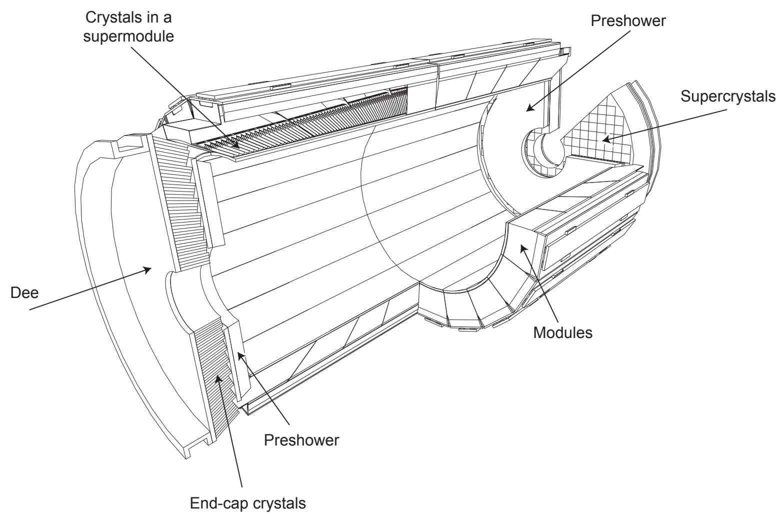


FIGURE 3.8: CMS ECAL layout

Lead tungstate is benefitted as it helps to make ECAL compact with high granularity due to the fact that it is denser (8.28 g/cm^3) with less radiation length (X$\circ$ of 0.89 cm). Also, lead tungstate has small Molière radius, which defines the area of deposited energy, helping to minimize the effect of pileup while measuring the energy. Other important factor is that crystals of lead tungstate have decay time of scintillation about 25 ns which is also the time of LHC bunch crossing.

ECAL operating temparature is around 18°C and light emitted from the scintillation process is temperature dependent. So, in order to maintain the temperature at nominal value, cooling system is installed which is based on water circulation through Aluminium pipes and is capable enough to maintain the temperature of detector upto $18 \pm 0.05°$ C. Layout of the ECAL of CMS detector is shown in the Figure. 3.8.

Energy resolution of ECAL, which is the ability of detecting system to accurately measure energy of incident radiation, can be written as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{A}{\sqrt{E}}\right)^2 + \left(\frac{\sigma_n}{E}\right)^2 + c^2 \tag{3.2}$$

where A is stochastic term (defines the intrinsic statistical fluctuations in shower, in energy deposits of preshower, and the photo(electron)-statistics contributions), σ_n is the electronics noise term and c is constant that covers the constant factors such as non-uniformity in detector for light collection and etc. Contribution of

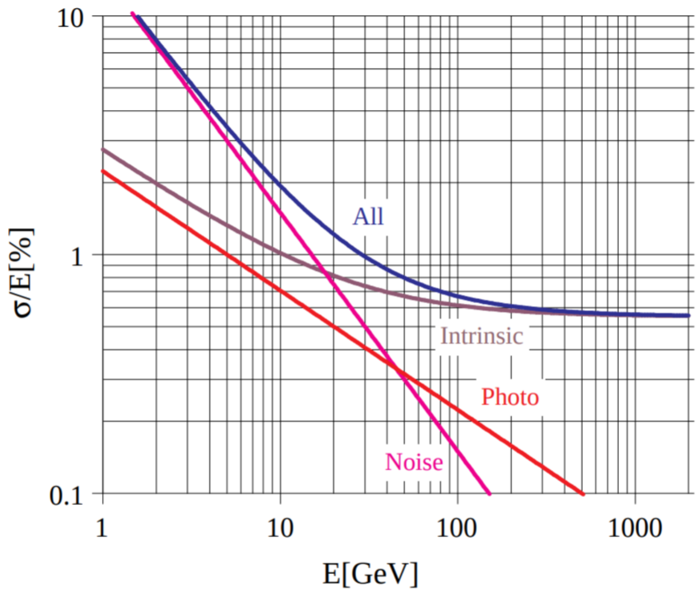


FIGURE 3.9: CMS ECAL energy resolution curve with individual term contributions.

each term to the energy resolution is shown in the Figure. 3.9.

A recent study on relative energy resolution of ECAL using 2017 data and MC has been performed using Z events decaying to a pair of electrons. The resolution plots for bremsstrahlung electrons (R_9>0.94 and R_9<0.94) is separately shown for data and MC in the Figure 3.10 [**Zghiche_2019**]. For H $\rightarrow 4\ell$ analyses, excellent energy resolution for leptons is important for precise measurements for which ECAL of CMS plays an important role as it did for Higgs discovery.

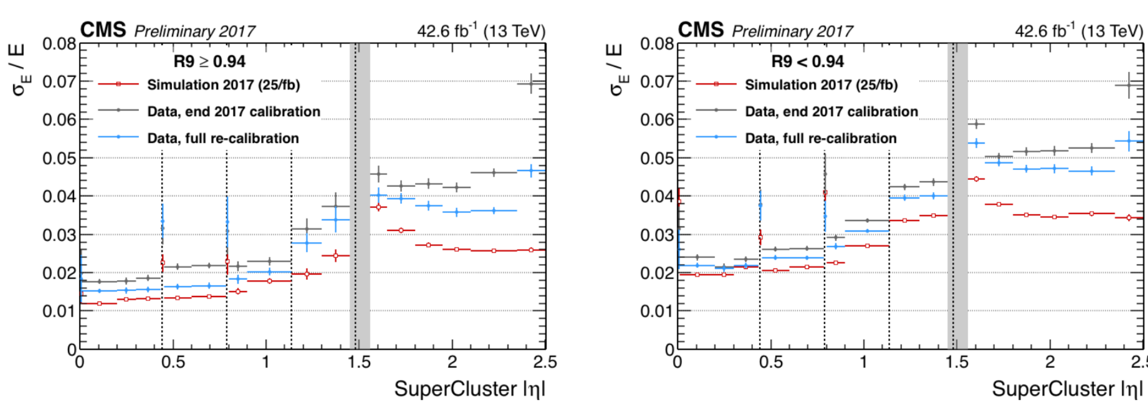


FIGURE 3.10: CMS ECAL energy resolution.

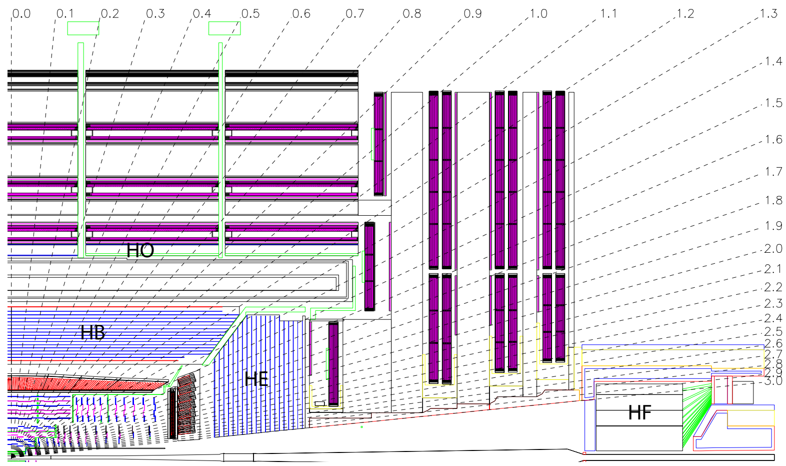


FIGURE 3.11: CMS HCAL layout.

3.1.5 Hadron Calorimeter (HCAL)

It is the part of CMS detecting system, which helps for the detection and as well as energy measurements of strongly interacting particles such as quarks (or the ones in composition of quarks), gluons hadronic jets (like neutrons, pions and etc.). Being hermetic, HCAL also helps to measure missing transverse energy (MET) which can be associated to invisible particles like neutrinos which is important studies like top quark production. HCAL assembles with CMS system between the solenoid magnet and ECAL as can be seen in the Fig. 3.1 at radial distance from 1.77 m to 2.95 m. Benefitting from pretty short interaction length, at most brass is chosen as absorber throughout the HCAL. Brass is a non-magnetic low cost material. HCAL provides pseudorapidity coverage upto $|\eta| < 3.0$ beyond which, 11.2 m away from the interaction point, froward Hadron Calorimeter (HF) is employed on both sides of the detector providing additional coverage upt $|\eta| <$ 5.2. HF helps to detect particles from proton-proton collisions emerging at very small angles with respect the beam axis.

Schematic diagram of HCAL is shown in the Figure 3.11 [20].

HCAL is splitted into barrel and endcap regions, HB and HE which give pseudorapidity coverage of $|\eta| < 1.33$ and HE of $1.3 < |\eta| < 3$ respectively.

HCAL is also splitted into barrel and encap regions, HB and HE respectively. HB consistes of 32 towers provides coverage of pseudorapidity of $|\eta| < 1.4$ and HE of $1.3 < |\eta| < 3$. This results in 2304 towers in a segmentation of $\Delta\eta \times \Delta\phi$ = 0.87 $\times 0.87$.

HB serves as a single longitudinal sampling readout. HB is split into two halves (HB^+ and HB^-) which consist of 36 similar wedges each weighing 26 tonnes made up of brass plates. It may happen that tail of showers of hadrons which leak from rear of calorimeters may penetrate into ECAL and/or magnet. Therefore, additional calorimeter called Hadron Outer (HO) is installed outside the solenoid that used the coil of the magnet as absorber. With this effective thickness of HCAL becomes above 10 interaction lengths. HO has scintillator of 10 mm thickness and helps to improve resolution of HCAL for MET.

Pseudorapidity coverage of $3 < |\eta| < 3$ is provided by sub part of HCAL whihc is called Hadron Forward (HF). It suffers huge flux of the particles due to which sampling material is required to be hard against the radiations. For this purpose, steel/quartz fibre was used. HF is located 11.2 away from interaction point with 1.65 m absorber depth. With the incident particle flux, Cerenkov radiations are emitted through quartz fibers which is then channeled to photomultipliers (PMs).

HCAL is constructed with alternate layers of plastic and brass (the absorber and scientillator respectively) which helps to measure the arrival time, position and energy of a particle. Plates of brass are 79 mm thick each with 99 mm gaps for the scintillators installation. When a particle passes through scintillator, optical fibers collect the emitted lights and pass it to readout box. i Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the from of 70,000 tiny tiles.

The energy resolution of the HCAL (i.e. HB, HO and HE) is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (a)^2 \tag{3.3}$$

and for that of HF it is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (b)^2. \tag{3.4}$$

Where a and b are constants having values 0.045 and 0.09 repectively.

There were subsequent upgrades in HCAL as in 2017, PMTs were replaced with new multi-anode PMTs and additionally electronics replaced for better differentiation of real hit from the muons which hit the PMT window. HE of HCAL was upgraded and photo detectors were replaced with latest silicon photo multipliers (SiPMs) for noise elimination and better photo detection efficiency [**pixel_hcal_upgrade**].

3.1.6 CMS Muon System

Identification of muons originating from the collision point is a great challenge. Having long lifetime (i.e. 2.19 $\times 10^{-6}$, which is even higher in relativitic regime), they tend to cross the whole detector without decaying. Also, having higher mass (about 200 times the mass of electrons) it tends to lose very little energy via bremsstrahlung. So, they leave track however they deposit very little amount of energy in the calorimeters. Hence a dedicated muon system was proposed and installed as outermost sub-detector of CMS [26]. The schematic diagram of CMS muon sytem (as stood by Run 2 operation of LHC) is shown in the Figure 3.12. It is benefitted from three different strategies for detection and measurements on muons.

- Drift Tubes (DT), only in barrel region
- Resistive plate chamber (RPC), in barrel and endcap regions both
- Cathode Strip Chamber (CSC), only in endcap region

The detection of the muons is very important for many physics processes particularly for the SM Higgs when it decays to ZZ/ZZ^*. Best mass resolution is obtained when each of Z decays to a pair of opposite sign muons which intereact weakly with the detector leaving their traces both in muoon system and as well other parts of the CMS detector. Due to this reason Higgs decaying to ZZ/ZZ^* is regarded as golden channel for Higgs properties measurements. Some theories beyond SM also require muons to be reconstructed and identified with ultimate precision which is obtained with the CMS muon system.

Muon system of CMS, benefitted from mainly gas detection technology, is designed in cylinderical shape that gives pseudorapidity coverage upto $|\eta| < 2.4$. RPC covers both barrel and endcaps, DT covers the barrel and CSC covers the endcap only. Overall goals of CMS muon system include muon identification, standalone and global muon resolution, charge assignment, triggering and capable of withstanding in high radiation and background enviroment.

Drift Tubes (DT)

Drift chambers play role of tracking detector in barrel region motivated to work under low event rate and relatively less strength of magnetic field.

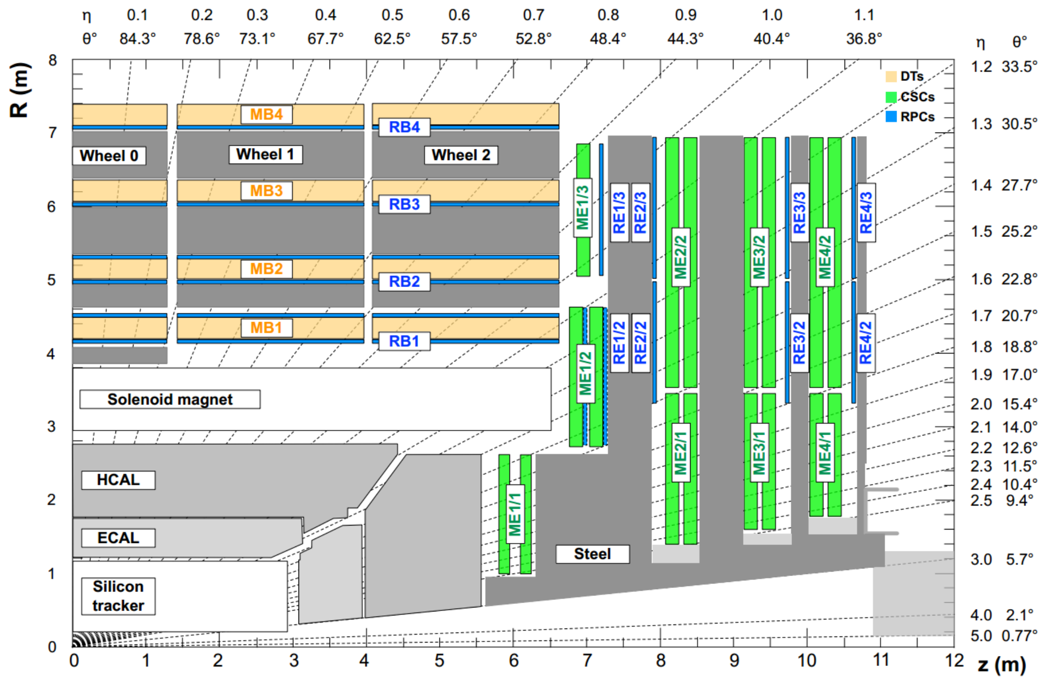
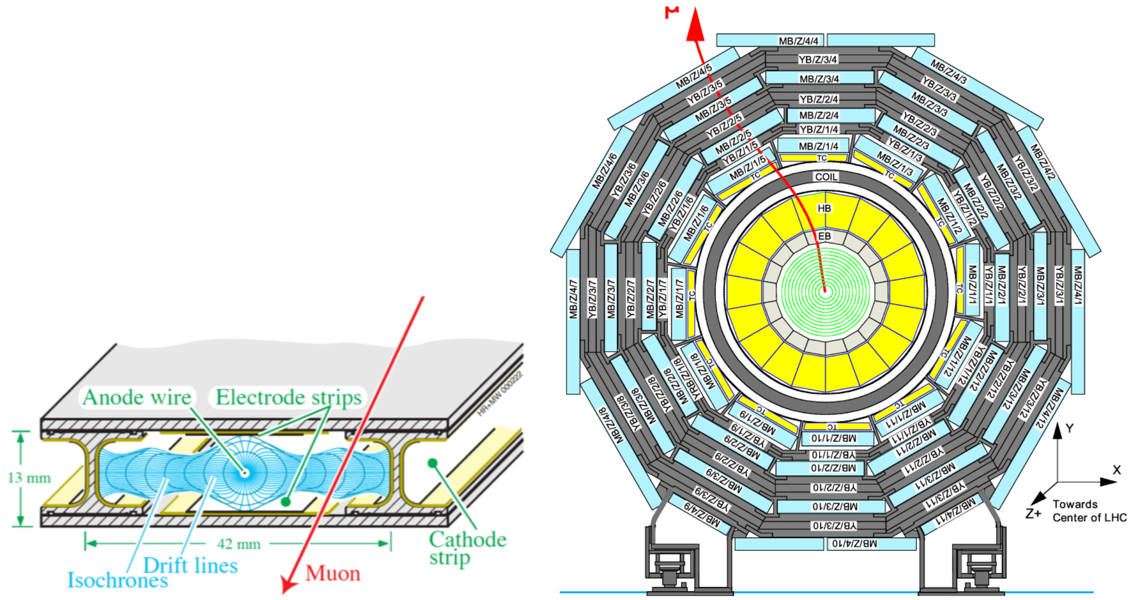


FIGURE 3.12: CMS muon system layout (as stood by Run2 operations of LHC).

The DTs make cover the barrel part of muon system alongwith the RPCs, as shown in the Figure. 3.12, providing pseudorapidity coverage of $|\eta| < 1.2$. DT system is annexed to the muon system in 4 stations, MB1, MB2, MB3 and MB4, in radially outward direction at 4.0, 4.9, 5.9 and 7.0 m respectively from IP. 250 DT chambers (composed of 12 Aluminium layers each) are fixed on 5 wheels along z-axis. 12 planes in a DT chamber (of average size 2m$\times$2.5m), except the ones from MB4, are build with 3 independent units which are called super-layer (SL). A set of two SLs measures coordinates in the bending plane (ϕ), the third SL helps to provide track coordinate along the beam (z). SL is further composed of DTs in 4 layers **[Ref->, cms_muon_tdr]**. Each DT cell has an anode wire made of steel (of radius 25 m) togther with walls of the cell as cathodes which produce the (drift) electric field. A DT cell is filled with mixture of CO_2 (15%) Ar (85%) gases which are benifitted from their low cost and provision of better drift velocity of saturation and the quenching under the influence of high operating voltage of 1.8kV. A schematic diagram of a drift cell and one of wheel bearing DT chambers is shown in the Figure . 3.13 and in Ref. **[comm, article_dt]** . As a charged muon moves throught a DT cell, it causes ionization in gas. The librated electrons then move towards anode under the influence of the electric field. Drift time of these electrons helps to determine distance between anode wire and the muon.



(a) (b)

FIGURE 3.13: (a) Schematic diagram of drift cell and produced electric field. (b) Cross sectional view of DT chambers in barrel shown one of the five wheels

To keep drift velocity steady, the filled gas is kept at a constant pressure inside the chambers. The pressure in each wheel is continuously monitored by sensors.

Resistive Plate Chambers (RPC)

RPCs, the parallel plate gaseous detectors, are important part of CMS muon system installed in both barrel and endcap region. RPCs are very fast responsive used for triggering and they offer good time and spatial resolutions. RPCs are capable to provide a time resolution of about a nano-second. A schematic diagram of a RPC chamber taken from **[article]** is shown in the Figure. 3.14.

RPCs are made of two high resistive plastic positive and negative charged plates, kept parallel to each other, with gas filled inn between them (95% of $C_2H_2F_4$ and 5% of C_4H_{10}). RPCs have readout system between two gaps and operate in the avalanche mode which is produced when muon passes through by librating electrons during primary and secondary ionization of molecules of gas. With the excellent time resolution, RPCs are capable to assign bunch crossing with ultimte precision.

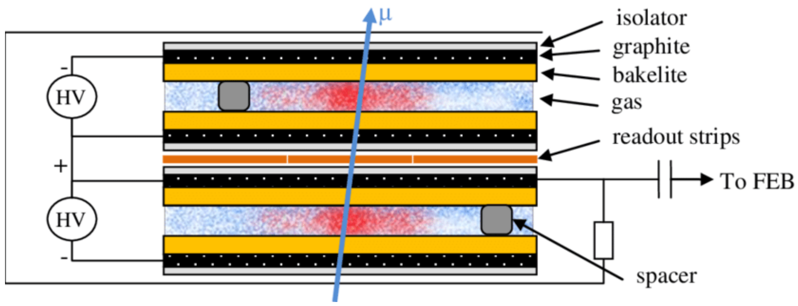


FIGURE 3.14: Cross sectional view of a RPC chamber.

Cathode Strip Chambers (CSC)

The cathode strip chambers (CSC) are the sub-detecting part of the CMS muon system, as shown in Figure **??**, covering the high pseudorapidty range upto $|0.9| < |\eta| < 2.4$ and capable to provide the track and triggering of muon in this region alongwith the RPCs. It can efficiently operate at high event rate and helps to assign the bunch crossing quite efficiently. A total 540 CSCs are installed in 4 CSC stations (ME1, ME2, ME3 and ME4) and each station is further distributed in 2-3 rings with 18-36 CSCs fixed on each ring.
Each CSC is assembly of 6 independent layers, making 6 gaps of gas (Ar 50%, 40% CO_2, 10% CF_4) with each layer have radial cathode strips and transverse anode wires. Muon passing through a CSC chambers ionizes the gas molecules thus producing the electron avalanche towards the anode wires made of copper. Produced gaseous ions which move towards the strips for the detection of image charge. These signals are collected by readout sytem, amplified and then passed to the electronic boards on chambers. In each chamber, muon hit information can be measured along the track of muon in the 6 layers. To generat a triggere primitive, hits in 4 layers out of 6 is required. Fitting the position of successive hits in the chambers gives the precision in muon momentum measurement. A layout of CSC and signal generation process is shown in the Figure. 3.15.

In a chamber layer, Cathode Front-End Boards (CFEBs) [1] receive and then process the signal from strips. However, from anode wires, signal is read by Anode Front-End Boards (AFEBs) and processby Anode Local Charged Track (ALCT)

[1] Electronics or readout system of ME/1/1 chambers were upgraded during long shutdown 1 (LS1), hence differs from other chambers.

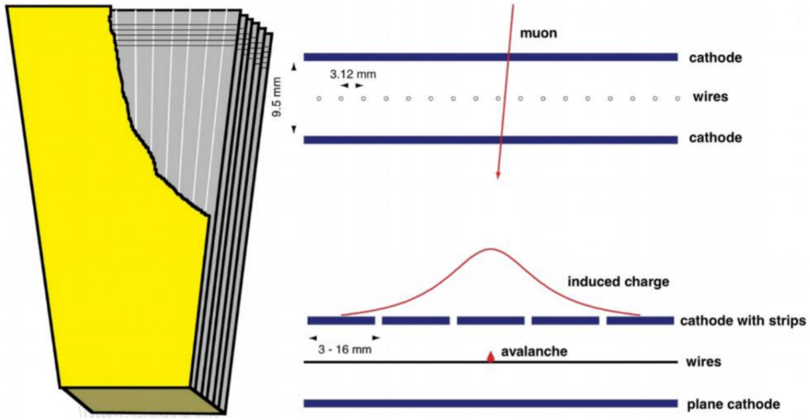


FIGURE 3.15: CSC layout (left) and induced current (right) when a charged particle pass through it.

board. These boards then communicate with the Trigger Mother Board (TMB) located in the vicinity of the endcap disks. Data is then transmitted to Data Mother Board (DMB) which sends it out Front End Driver (FET) crates whose front end electronics has interface with CMS data acquisition system. Overview of the chamber electronics and communication mechanism is elaborated in Figure. 3.16.

Improvements in CMS muon system during Run1 - Run2 and during ongoing long shutdowon 2

From Run1 to Run2 operations of LHC, increasing luminosity was challenging for muon system of CMS. In order to cope with high event rate environment and utilize the gain in the luminosity, muon system of CMS underwent some upgrades or modifications [**Sirunyan:2018fpa**]. To minimize the misidentification rates, and improve efficiency and redundancy, RPC and CSC were extended to install RE4 and ME4 as fourth stations in the mainstream. Due to this, CSC and RPC Trigger Track-Finder were rebuilt to accommodate the additional information from ME4 and RE4 planes. Readout electronics and trigger system were also enhanced. To increase the bandwidth, optical fibre links in DT and CSC were added. During Run1, strips in ME1/1a ring were ganged together into a group of 3. Subsequently, new electronics, Digital Cathode Front End Boards (DCFEBS), were installed in ME1/1 chambers of CSC to provide readout to every strip separately allowing ME1/1 to effectively perform in high luminosity environment.

During ongoing long shutdown 2 (LS2), there several upgrades are in process for

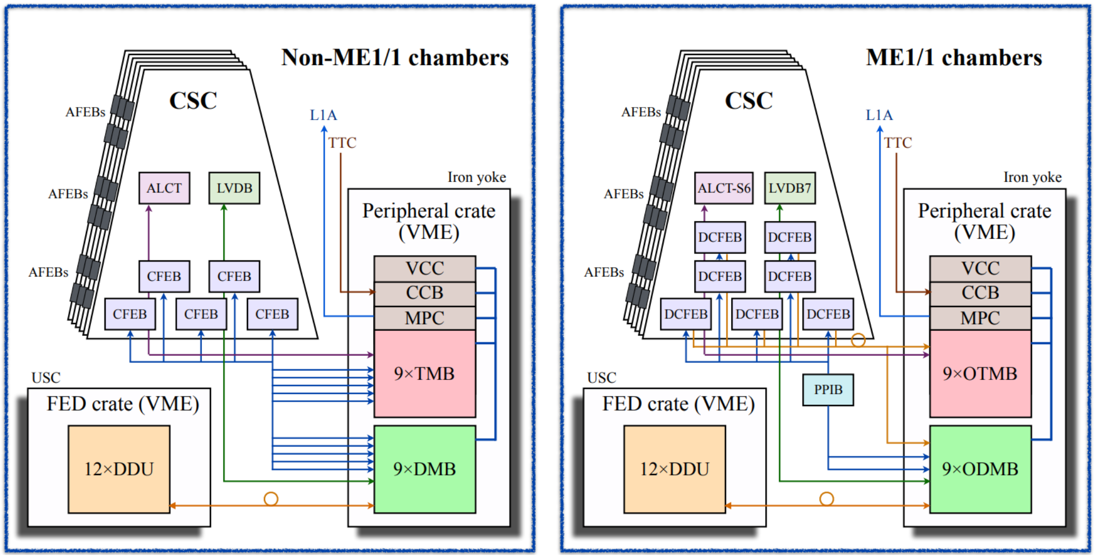


FIGURE 3.16: Overview of electronic boards of a CSC chamber and involved in communication of data. Readout system of ME/1/1 chambers were upgraded after Run 1, hence differs from other chambers.

upcoming physics programs under high luminosity LHC. CSC will undergo upgrade of readout system of ME1/1, ME2/1, ME3/1, and ME4/1. Total 180 chambers will be refurbished to replace CFEBs with upgraded DCFEBs and ALCT mezzanine boards. Also for ME2/1, ME3/1 and ME4/1, LVDBs will also be upgraded.

Other main improvement for the muon sytem for future programs is installation of Gas Electron Multiplier (GEM) chambers in forward regions. Due to upcoming higher collision rate during Run 3 and so on, muon system needs to improve its capibilities for tracking and the triggering to effidiently work extreme environment which can be achieved with GEM chambers. In this program, first two muon stations will be supported with GE1/1 and GE2/1 (covering $1.5 < |\eta| < 2.4$) which have layers for high precision measurement. Additionally, ME0 station will be installed to extend coverage of muon system upto $|\eta| < 3$. GE1/1 among these, will be installed during LS2 and would take part in Run 3 operations; however, GE1/2 and ME0 would be installed after Run 3 operations during long shutdown 3 (LS3) after 2022 when RPC will also upgraded as “Improved RPC (iRPC)" with new geometry, materials and frontend electronics. Updated schematic diagram of CMS muon system with addition of GEM chambers can be seen in the Figure. **??** (also in Ref. [**Radogna:2016esa**, **Costantini:2711362**]) where the actual position of

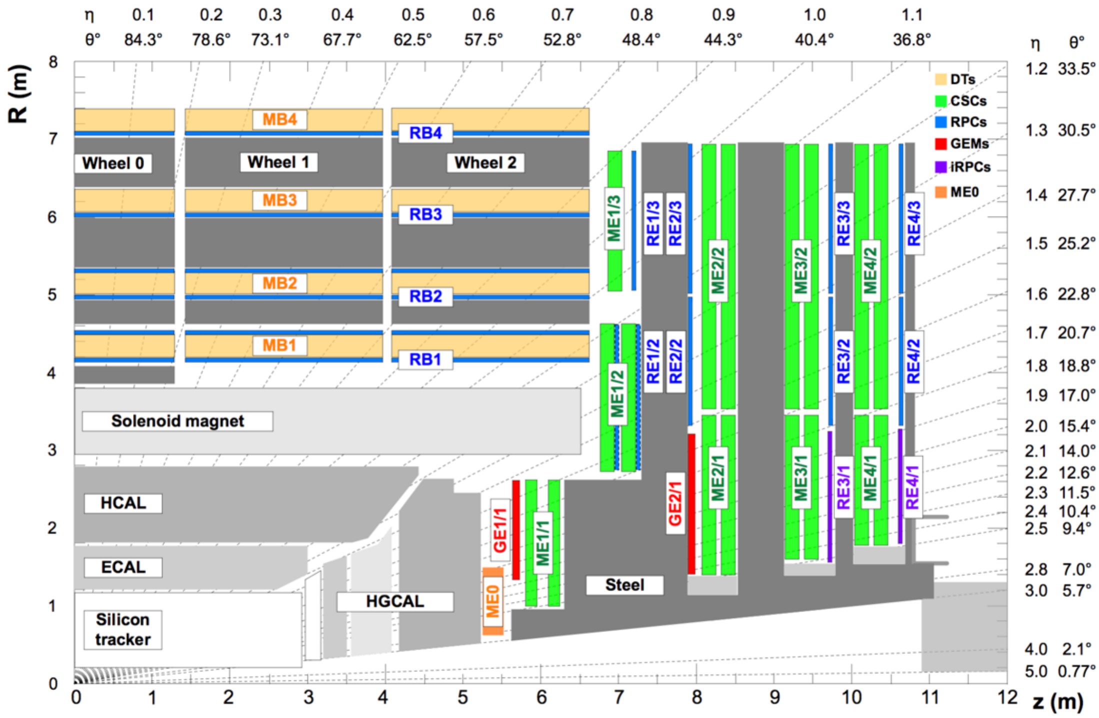


FIGURE 3.17: Updated schematic diagram of CMS muon system with addition of GEM chambers.

these chambers in muon system can be understood.

3.2 CMS Trigger System

CMS is big and general purpose experiment purposed to explore data that LHC has produced during the proton-proton collisions. LHC produces around 40 million collisions in a second (40 MHz); however, not all the collisions are interesting for physics purposes. Further, for the data acquisition (DAQ) system, it is not effective to read every event produce at high rate. Also, storing the uninteresting eventsprouced in LHC collisions will require a lot of storage. Hence, there is a dire need to reduce the rate of produced events during LHC collisions which is done by the trigger system which select only those events which are crucial for physics purposes. Trigger system of CMS is a sophisticated two-level system; Level 1 (L1) trigger and high level trigger (HLT).

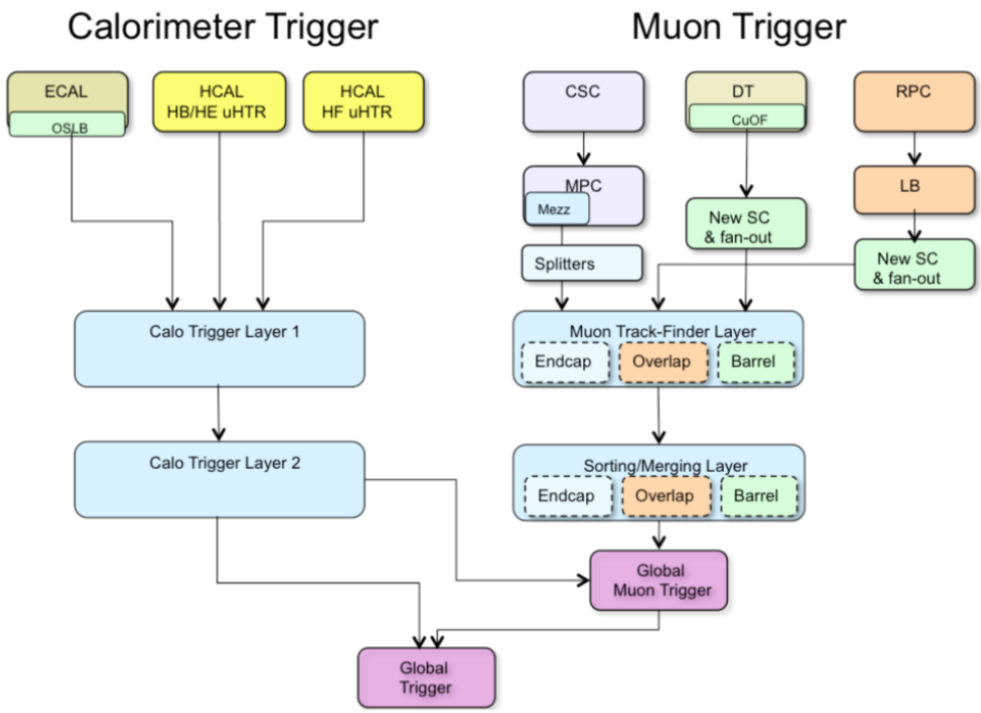


FIGURE 3.18: The configuration of L1 trigger.

3.2.1 The L1 Trigger of CMS

The L1 trigger uses CMS hardware and taking informations from the calorimeters and the CMS muon system to make decisions (in 4 s) in several steps to accept or reject an event. L1 trigger reduces event rate from 40 MHz to 100 kHz. A sketch of L1 trigger is shown in the Figure. 3.18.

L1 trigger is build with programmable electronics, that exploits coarse information of energy (ECAL and HCAL) and momentum (DT, RPC and CSC) of electrons, muons, jets, photon and MET. L1 trigger can be classified into calorimeter and muon triggers. Calorimeter has ECAL and HCAL as independent regional triggers that provide information to the global Calorimeter Trigger (GCT). Likewise, muon trigger has local trigger corresponding to CSC, DT and RPC each which then combine to make global muon trigger (GMT). GCT and GMT then feed to Global Trigger (GT) to make final decision about the event whether to accept or reject.

In order to cope with higher event rate for Run 2 operations, a new trigger architecture, the Time Multiplexed Trigger (TMT) was planted. TMT uses sophisticated algorithms (i.e. FPGAs) in μTCA to exploit full granularity of the calorimeters and is benefitted from technological improvements in hardware. rebuilding the CSC Trigger Track-Finder to accommodate the additional information from ME4/2 and ME1/1

High-Level Trigger System (HLT)

HLT performs event selections in a similar way that we perform for the offline processing of events. Primary goal of the HLT system is to take events (coming at 100kHz) from L1 trigger and reduce it to get an output rate of 1 kHz. This is done is done in several steps. For a process of interest, to reconstruct or identify an object (like electron, muon or so) a certain criteria is applied over each event. HLT is built with a farm of single processor computers which is known as event filter farm which assembles fragments of an individual event with the help of the builder units to construct the complete events. An assembled event is then fed to the filter unit which transforms the raw information into the data structure which is specific to the detector and does event reconstruction and filterig of trigger. These builder and filter units in a machine and linked through a shared memory. Filtering process exploits the precise data coming out of the trigger wherein event selection is based on the quality of the offline reconstruction algorithms. Available CPU power allows to take events at rate of 100 kHz which sustains for average processing time per event viz 90 ms which can be increased with an increased CPU power.

Performance of HLT is based on *HLT paths* which consist of a set of algorithms to perform in a predefined order to perform physics object selections followed by reconstruction step. Each HLT path applies in growing order of complexity as simpler algorithms run first to impose basic selections. Different HLT paths are based on diffrent algorithms which are designed quite similar to ones used for offline reconstruction in analysis. This helps to use CMS software which is used for offline analyses.

Events passing L1 trigger are sent to HLT menu having certain paths which are assigned to make decision. HLT operates at a specific instantaneous luminosity which is done by dividing the passed events with a paramer, n, called prescale factor. HLT path with pre-scale value n selects one event out of n passing events. Events organized in streams of HLT decisions are then passed to storage manager

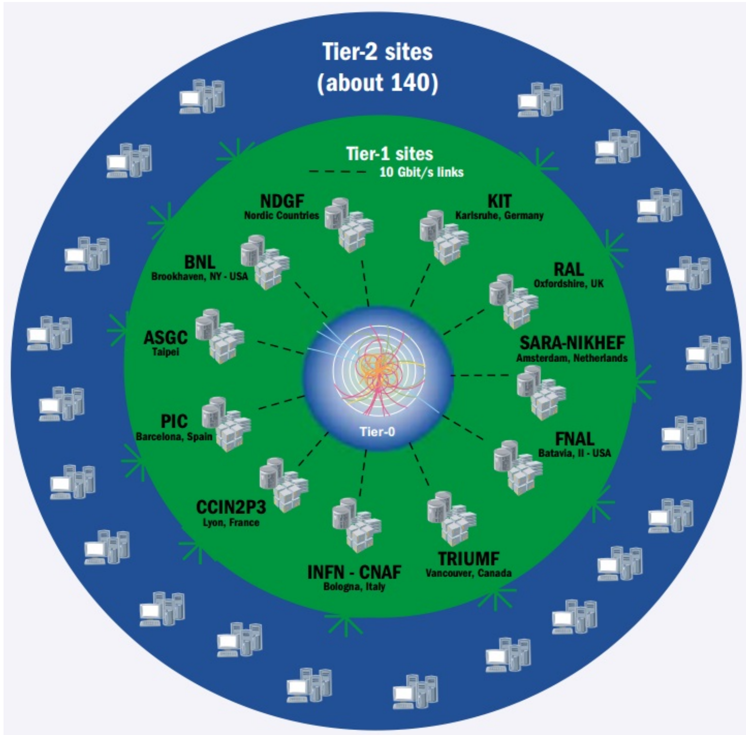


FIGURE 3.19: A sketch of WLCG Tier architecture.

softwares which store them on local disks and finally transfer to Tier-0 of CMS for its permanent storage.

3.3 Worldwide LHC Computing Grid

There is a dire demand of the four experiments of LHC (CMS, ATLAS, ALICE, LHCb) for a facility to store the prompt data obtained from LHC collisions. Further to process and analyse the huge amount of data to extract physics results is also challenging. LHC produces about 15 petabyte data each year duing its operation. To cope with this huge amount of data, setting a multi-tiered computing infrastructure was approved by CERN council in 2002. This is known as Worldwide LHC Computing Grid (WLCG) and which links around 170 computing centers in more than 40 countries. WLCG is mainly tasked for providing infrastructure of organized computing facilities to the four main experiments of CERN. To fulfill the goals, it is connected with around 150 thousand cores of processors with storage capicity of 50 petabytes on the disks **[Fisk:2010jd]**.

Schematic diagram of WLCG Tier system is shown in the Figure 3.19.

3.3.1 3-Tier Architecture

Computing models at WLCG are mainly based on 3 tier levels. Tier-0 facility is situated at CERN. It receives prompt data (RAW data) produced from LHC collisions as a primary backup and does the reconstruction of the events and store in the form of prompt-RECO dta. It then exports the RAW data to to large computing centers at Tier-1 where the events are set to be reconstructed again with refined calibrations and at each Tier-1 site, a detailed reconstruction of RAW data is performed to get AOD(Analysis Object Data) version of data which most compact form of data. It also serves to distribute the AOD data to Tier-2 sites where the data can be used for end-user analysis specific reprocessing and simulations for all experiments.

Chapter 4

Summary and Outlook

4.1 Summary of the dissertation

After many years of search, ATLAS and CMS collaborations of CERN announced the discovery much awaited Higgs boson in early July 2012 verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation. With availble LHC data, measured properties of Higgs are consistent with SM. Inclusive and differential cross section measurements of Higgs are its very important properties which help to probe its structure and its nature of interaction with decaying particles. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

Measurement of inclusive and differential cross section of the Higgs boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV corresponding to an integrated luminosity of 137.1 $\mathrm{fb}^{-1}$is presented in this dissertation. Measurements are performed by exploiting $H \to 4\ell$ decay channel ($\ell = e, \mu$), which is known to be a golden channel for such measurements. To minimize the dependence of the measurements on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial volume that closely matches with experimental acceptance. Remaining model dependence is extracted by repeating the measurement for SM and other exotic model and reported as a distinct systematic uncertainty. All measurements include the correction for the finite detection efficiency and resolution effects to allow for direct comparison with the SM predictions. As $\sigma \times BR$ strongly depend upon Higgs mass, the measurements have been extracted assuming $m_H = 125.09$ GeV, as reported by CMS experiment [27] from combined measurement in the $H \to 2\gamma$ and $H \to 4\ell$ decay channels.

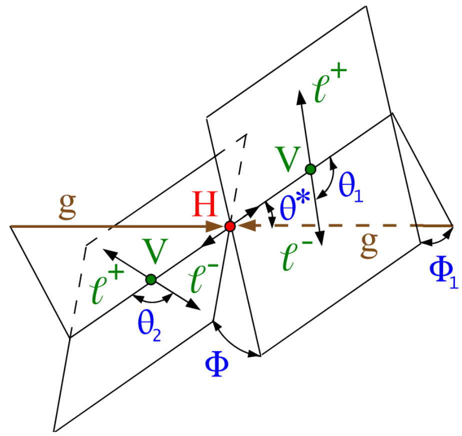


FIGURE 4.1: Higgs event signature

The differential fiducial cross sections are presented as a function of kinematic observables which are sensitive to Higgs production mechanism: rapidity and transverse momentum of Higgs, associated jet multiplicity and transverse momentum of the leading jet. Full Run 2 data (137.1 fb^{-1}) enabled us to explore the differential observables in more fine bins than presented in previous measurements on this topic.

The integrated fiducial cross section is measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})$ $^{+0.24}_{-0.19}(\mathrm{syst.})\ \mathrm{fb}$ at 13 TeVand SM predicts this to be $2.76^{+0.14}_{-0.14}$. The agreement between SM predictions and measured $H \rightarrow 4\ell$ cross section measurement at $\sqrt{s}$ =13 TeV is found be good. These model independent cross section measurements can be compared with any theoretical predictions in future with better accuracy.

4.2 Outlook

The presented measurements of Higgs differential cross sections are related to Higgs production. There are certain things that can be done which are beyond the scope of this dissertation:

- With more data, get more precise results
- Study more differential observables
- Include additional theoretical prediction(s) for comparison with the observed cross section in differential bins

- Include potential theory interpretations for differential observables

The current measurements are emerged with significant statistical uncertainties. The studies based on simulations show that the uncertainties on Higgs cross section measurements with 300 fb^{-1}pp collisions data (as expected from LHC Run 3) would be significantly less than the current theoretical uncertainties. More data will help to affirm the SM Higgs boson if measurements be in agreement with the SM expectations, elsewise any deviation from SM predictions will open the door to some new physics.

Studying more differential observables are also important from different aspects. There are more *production* and as well as *decay* related observables ($\theta^*, \theta_1, \theta_2, \Phi, \Phi_1$ as shown in Figure. 4.1) which can be studied with currently available data from CMS [**event_signature**]. With respect to earlier iterations, current data would allow to exploit more differential bins for a particular observable. Moreover, with more statistics there is opportunity to study the double differential cross section (as function of two observables). In $H \to 4\ell$, there are ongoing efforts for additional differential measurements alongwith potential theory interpretations; however, it is yet to accomplish to be made public. Preliminary version of some of the new differential measurements have been shown in Appendix. In the reported measurements, the observed cross sections are compared with respective theoretical predictions based on POWHEG and NNLOPS; however, including additional theoretical prediction(s) e.g. MADGRAPH5 are also underway to be part of next manuscript.

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